

Photoplethysmography (PPG) measures the heart rate based on the absorption of visible light in the body. An LED illuminates the tissue and a light detector monitors absorption, as shown in Figure 1. The tissue blood volume (B_V) has a large influence on the light absorption. B_V increases when the heart contracts and decreases when the heart relaxes, causing the PPG signal to increase and decrease in response to the changes in B_V .

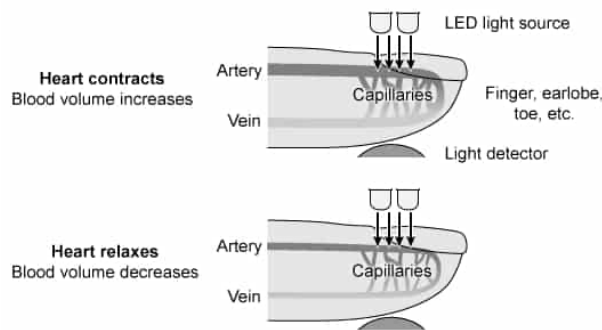


Figure 1 Diagram of a PPG system during the cycle of heart contractions and relaxations

The LED circuit consists of a voltage source, a current-limiting resistor, and the LED connected in series. The LED itself does not behave like a resistor: When current flows through the LED, the LED voltage remains constant, known as the forward voltage. The LED cannot handle large currents, so the current-limiting resistor keeps the current in the circuit below the maximum forward current of the LED.

The PPG signal shows when blood flows into the tissue, but an ECG indicates heart contractions. Figure 2 compares these signals, revealing a time delay that depends on the pulse wave velocity, which is the rate the pressure wave moves through the vasculature. Healthy blood vessels dilate when blood pressure increases, slowing the pulse wave as it leaves the heart. Increasing age and some diseases cause "hardening" of the blood vessels, increasing the pulse wave velocity.

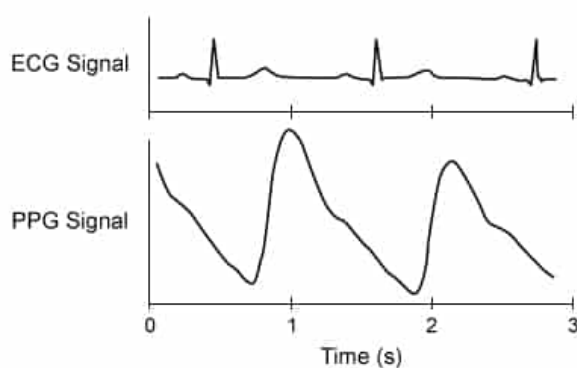


Figure 2 Simultaneous ECG signal and PPG signal showing the effects of the pulse wave velocity

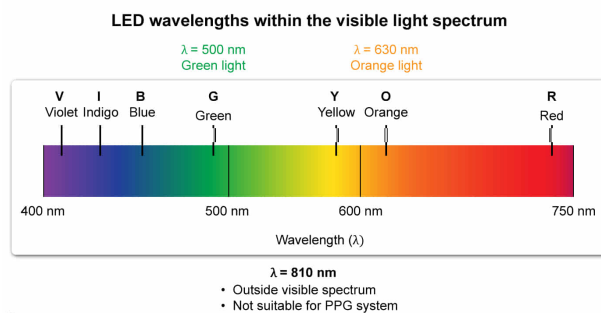
Which of the following LED wavelengths would be suitable for use in a PPG system?

- I. 500 nm
 - II. 630 nm
 - III. 810 nm
- ☐ A. I only [4%]
- ✓ ☒ B. I and II only [81%]
- ☐ C. II and III only [10%]
- ☐ D. III only [3%]

Correct

80% answered correctly

Explanation:



In this question, the PPG system uses an LED that produces **visible light**. Visible light consists of the **wavelengths** from about 400 nm (violet) to about 750 nm (red). An LED that produces a wavelength of light outside this range would not be suitable for use in a PPG system. LEDs with wavelengths of 500 nm and 630 nm are within the visible light spectrum (**Numbers I and II**).

(Number III) 810 nm light is outside the visible light spectrum and would not be suitable for use in a PPG system.

Educational objective:

The visible light spectrum includes wavelengths from about 400 nm to about 750 nm. LEDs that produce light within this range are suitable for PPG applications.

Time spent: 0 sec

Subject:
Physics

QID: 403789

System:
Electrostatics and Circuits